

Experiment 1: Fixed Height + Fixed Wind + No Obstacle | fixed_h_fixed_w_no_obs | Comparison Table

Dropped-Only Quality

Method	N(drop)	Precision Mean(m)	Precision P50(m)	Precision P90(m)	R@1m	R@2m	R@3m	Roll Mean	Roll P90	Pitch Mean	Pitch P90	Impact Mean	Impact P90	Impact P95
PPO	1984	2.304	1.313	4.804	36.7%	71.5%	83.3%	14.198	28.817	8.774	22.902	0.567	1.295	1.434
PPO-GRU	1988	2.976	0.779	13.882	62.5%	82.6%	84.7%	7.712	17.813	6.125	13.707	0.351	0.784	1.029
PPO-RNN	1987	2.171	1.164	3.856	41.8%	75.6%	87.2%	15.917	37.581	6.916	14.208	0.484	1.305	1.476
MPC	1053	6.026	3.100	16.294	3.6%	16.6%	46.1%	2.991	3.705	1.655	2.773	0.165	0.145	0.163

No-Drop Breakdown (done but no release)

Method	N(done)	N(drop)	Drop Rate	N(no_drop)	No-drop Rate	Crash Rate	Crash+NoDrop	Crash in NoDrop	Timeout+NoDrop	Both Flags
PPO	2000	1984	99.2%	16	0.8%	0.8%	16	100.0%	0	0
PPO-GRU	2000	1988	99.4%	12	0.6%	0.6%	12	100.0%	0	0
PPO-RNN	2000	1987	99.3%	13	0.7%	0.7%	13	100.0%	0	0
MPC	2000	1053	52.6%	947	47.4%	47.4%	947	100.0%	0	0

Cross-Case Check: PPO/PPO-GRU/PPO-RNN on MPC No-Drop Cases (large error > 3.0m)

Method	In MPC-NoDrop	Dropped	Drop Rate	Err>3.0m	LargeErr Rate (in dropped)	Mean Err (in dropped)
PPO	947	935	98.7%	189	20.2%	2.482
PPO-GRU	947	938	99.0%	123	13.1%	2.682
PPO-RNN	947	939	99.2%	116	12.4%	2.208